Reproduction

Biology in essence is the story of life on earth. While individual organisms die without fail, species continue to live through millions of years unless threatened by natural or anthropogenic extinction.

Why reproduction matters

- reproduction becomes a vital process without which species cannot survive for long
- each individual leaves its progeny by asexual or sexual means
- the sexual mode of reproduction enables the creation of new variants, so that survival advantage is enhanced

What this unit covers

- the details of reproductive processes in flowering plants and in humans, as easy to relate representative examples
- a related perspective on human reproductive health, and how reproductive ill health can be avoided
- together these complete our understanding of the biology of reproduction

In this unit

Ch 1 Sexual Reproduction in flowering Plants

Ch 2 Human Reproduction

Ch 3 Reproductive Health

Panchanan Maheshwari, 1904 to 1966

- born in November 1904 in Jaipur, Rajasthan, he rose to become one of the most distinguished botanists not only of India but of the entire world
- he moved to Allahabad for higher education, where he obtained his D.Sc.
- during his college days he was inspired by Dr W. Dudgeon, an American missionary teacher, to develop an interest in Botany and especially morphology
- his teacher once said that if his student progressed ahead of him it would give him great satisfaction, and these words encouraged Panchanan to enquire what he could do for his teacher in return

The work

- he worked on embryological aspects and popularised the use of embryological characters in taxonomy
- he established the Department of Botany, University of Delhi as an important centre of research in embryology and tissue culture
- he emphasised the need for initiation of work on artificial culture of immature embryos, and these days tissue culture has become a landmark in science
- his work on test tube fertilisation and intra-ovarian pollination won worldwide acclaim

The honours, and the teaching

- he was honoured with fellowship of the Royal Society of London (FRS), of the Indian National Science Academy and of several other institutions of excellence
- he encouraged general education
- he made a significant contribution to school education by his leadership in bringing out the very first textbooks of Biology for Higher Secondary Schools published by NCERT in 1964

Why flowers are as they are

- the myriads of flowers that we enjoy gazing at, the scents and the perfumes that we swoon over and the rich colours that attract us are all there as an aid to sexual reproduction
- flowers do not exist only for us to be used for our own selfishness
- all flowering plants show sexual reproduction
- a look at the diversity of structures of the inflorescences, flowers and floral parts shows an amazing range of adaptations to ensure formation of the end products of sexual reproduction, the fruits and seeds

1.1 The flower, an organ of angiosperms

What this chapter sets out to do

- to understand the morphology, the structure and the processes of sexual reproduction in flowering plants, that is angiosperms

Flowers and human beings

- human beings have had an intimate relationship with flowers since time immemorial
- flowers are objects of aesthetic, ornamental, social, religious and cultural value
- they have always been used as symbols for conveying important human feelings such as love, affection, happiness, grief and mourning
- listing five flowers of ornamental value commonly cultivated at homes and in gardens, and five more used in social and cultural celebrations, is worth doing, as is finding out what floriculture refers to

Calyx of sepals and corolla of petals, the outer whorls

Androecium, the whorl of stamens

Gynoecium, with stigma, style and ovary

Thalamus, on which the whorls are borne

Ovule within the ovary, seen in longitudinal section

Figure 1.A A diagrammatic representation of L.S. of a flower

The flower to a biologist

- flowers are morphological and embryological marvels and the sites of sexual reproduction
- the two parts of a flower in which the two most important units of sexual reproduction develop are worth naming from the recalled parts of a typical flower

1.2 Pre-fertilisation structures and events

Before the flower is seen

- much before the actual flower is seen on a plant, the decision that the plant is going to flower has taken place
- several hormonal and structural changes are initiated which lead to the differentiation and further development of the floral primordium
- inflorescences are formed which bear the floral buds and then the flowers

The two reproductive whorls

- in the flower the male and female reproductive structures, the androecium and the gynoecium, differentiate and develop
- the androecium consists of a whorl of stamens, representing the male reproductive organ
- the gynoecium represents the female reproductive organ

Filament, the long and slender stalk

Anther, the terminal generally bilobed structure

Two theca in each lobe, hence dithecous

Four microsporangia at the corners, two in each lobe

Figure 1.B (a) A typical stamen; (b) three–dimensional cut section of an anther

1.3 Stamen and microsporangium

The stamen

- a typical stamen has two parts, the long and slender stalk called the filament and the terminal generally bilobed structure called the anther
- the proximal end of the filament is attached to the thalamus or the petal of the flower
- the number and length of stamens are variable in flowers of different species, so collecting a stamen each from ten flowers of different species and arranging them on a slide shows the large variation in size seen in nature, while careful observation under a dissecting microscope and neat diagrams elucidate the range in shape and attachment of anthers

The anther in section

- a typical angiosperm anther is bilobed with each lobe having two theca, that is, it is dithecous, and often a longitudinal groove runs lengthwise separating the theca
- the bilobed nature of an anther is very distinct in transverse section
- the anther is a four-sided, that is tetragonal, structure consisting of four microsporangia located at the corners, two in each lobe
- the microsporangia develop further and become pollen sacs, extending longitudinally all through the length of an anther and packed with pollen grains

Structure of a microsporangium

- in transverse section a typical microsporangium appears near circular in outline, and is generally surrounded by four wall layers, the epidermis, the endothecium, the middle layers and the tapetum
- the outer three wall layers perform the function of protection and help in dehiscence of the anther to release the pollen
- the innermost wall layer is the tapetum, which nourishes the developing pollen grains, its cells possessing dense cytoplasm and generally having more than one nucleus, which leaves the way tapetal cells become bi-nucleate to be worked out
- when the anther is young, a group of compactly arranged homogenous cells called the sporogenous tissue occupies the centre of each microsporangium

Transverse section of a young anther, its four corners

One microsporangium enlarged, showing the wall layers

Epidermis, endothecium, middle layers and tapetum

A mature dehisced anther, releasing its pollen

Figure 1.C (a) Transverse section of a young anther; (b) Enlarged view of one microsporangium showing wall layers; (c) A mature dehisced anther

1.4 Microsporogenesis

From sporogenous tissue to pollen

- as the anther develops, the cells of the sporogenous tissue undergo meiotic divisions to form microspore tetrads, which leaves the ploidy of the cells of the tetrad to be worked out
- each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad, so each one is a potential pollen or microspore mother cell
- the process of formation of microspores from a pollen mother cell (PMC) through meiosis is called microsporogenesis, and the microspores as they are formed are arranged in a cluster of four cells, the microspore tetrad
- as the anthers mature and dehydrate the microspores dissociate from each other and develop into pollen grains, several thousands of them being formed inside each microsporangium and released with the dehiscence of the anther

1.5 The pollen grain

The male gametophyte

- the pollen grains represent the male gametophytes
- touching the opened anthers of Hibiscus or any other flower leaves a deposition of yellowish powdery pollen grains on the fingers
- sprinkling these grains on a drop of water on a glass slide and observing under a microscope shows the variety of architecture, the sizes, shapes, colours and designs seen on the pollen grains from different species

Scanning electron micrographs of pollen grains

The array of patterns and designs on the exine

Sizes, shapes and surface sculpturing across species

Figure 1.D Scanning electron micrographs of a few pollen grains

A pollen grain tetrad, the four cells still together

Microspore, the single freed cell

Vegetative cell, larger, with an irregular nucleus

Generative cell, small and spindle shaped within it

Figure 1.E (a) Enlarged view of a pollen grain tetrad; (b) stages of a microspore maturing into a pollen grain

The two walls

- pollen grains are generally spherical, measuring about 25-50 micrometers in diameter, and have a prominent two-layered wall
- the hard outer layer, the exine, is made up of sporopollenin, one of the most resistant organic materials known, able to withstand high temperatures and strong acids and alkali, and no enzyme that degrades it is so far known
- the exine has prominent apertures called germ pores where sporopollenin is absent, and pollen grains are well-preserved as fossils because of sporopollenin, which leaves the reason the exine should be hard, and the function of the germ pore, to be worked out
- the inner wall is the intine, a thin and continuous layer made up of cellulose and pectin, and the cytoplasm of the pollen grain is surrounded by a plasma membrane

The cells inside

- when the pollen grain is mature it contains two cells, the vegetative cell and the generative cell
- the vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus
- the generative cell is small and floats in the cytoplasm of the vegetative cell, spindle shaped with dense cytoplasm and a nucleus
- in over 60 per cent of angiosperms pollen grains are shed at this 2-celled stage, while in the remaining species the generative cell divides mitotically to give rise to the two male gametes before the pollen grains are shed, that is a 3-celled stage

Pollen allergy

- pollen grains of many species cause severe allergies and bronchial afflictions in some people, often leading to chronic respiratory disorders such as asthma and bronchitis
- Parthenium or carrot grass, which came into India as a contaminant with imported wheat, has become ubiquitous in occurrence
- it causes pollen allergy

Pollen tablets and syrups, sold as food supplements

Claimed to increase the performance of athletes

And of race horses

Figure 1.F Pollen products

1.6 Pollen as food, and pollen viability

Pollen products

- pollen grains are rich in nutrients, and it has become a fashion in recent years to use pollen tablets as food supplements
- in western countries a large number of pollen products in the form of tablets and syrups are available in the market
- pollen consumption has been claimed to increase the performance of athletes and race horses

How long pollen stays viable

- once shed, pollen grains have to land on the stigma before they lose viability if they are to bring about fertilisation
- the period for which pollen grains remain viable is highly variable and depends to some extent on the prevailing temperature and humidity
- in some cereals such as rice and wheat pollen grains lose viability within 30 minutes of their release, while in some members of Rosaceae, Leguminoseae and Solanaceae they maintain viability for months
- it is possible to store pollen grains of a large number of species for years in liquid nitrogen at -196 degrees C, and such stored pollen can be used as pollen banks, similar to seed banks, in crop breeding programmes

1.7 The pistil

One pistil or many

- the gynoecium represents the female reproductive part of the flower, and may consist of a single pistil, that is monocarpellary, or of more than one pistil, that is multicarpellary
- when there is more than one, the pistils may be fused together, that is syncarpous, or may be free, that is apocarpous
- each pistil has three parts, the stigma, which serves as a landing platform for pollen grains, the style, the elongated slender part beneath it, and the ovary, the basal bulged part
- inside the ovary is the ovarian cavity or locule, and the placenta is located inside it, so the definition and types of placentation studied in Class XI are worth recalling

Hibiscus flower dissected to show the pistil

Papaver, a multicarpellary syncarpous pistil

Michelia, a multicarpellary apocarpous gynoecium

An anatropous ovule, with funicle, hilum, integuments, micropyle and chalaza

Figure 1.G (a) A dissected flower of Hibiscus showing pistil (other floral parts have been removed); (b) Multicarpellary, syncarpous pistil of Papaver ; (c) A multicarpellary, apocarpous gynoecium of Michelia; (d) A diagrammatic view of a typical anatropous ovule

1.8 The megasporangium, or ovule

Where ovules arise

- arising from the placenta are the megasporangia, commonly called ovules
- the number of ovules in an ovary may be one, as in wheat, paddy and mango, or many, as in papaya, water melon and orchids

The parts of an ovule

- the ovule is a small structure attached to the placenta by means of a stalk called the funicle, and the body of the ovule fuses with the funicle in the region called the hilum, so the hilum represents the junction between ovule and funicle
- each ovule has one or two protective envelopes called integuments, which encircle the nucellus except at the tip, where a small opening called the micropyle is organised
- opposite the micropylar end is the chalaza, representing the basal part of the ovule
- enclosed within the integuments is a mass of cells called the nucellus, whose cells have abundant reserve food materials, and located in the nucellus is the embryo sac or female gametophyte, an ovule generally having a single embryo sac formed from a megaspore

1.9 Megasporogenesis

The megaspore mother cell

- the process of formation of megaspores from the megaspore mother cell is called megasporogenesis
- ovules generally differentiate a single megaspore mother cell (MMC) in the micropylar region of the nucellus
- it is a large cell containing dense cytoplasm and a prominent nucleus

Megaspore mother cell, large, in the nucellus

Dyad and tetrad of megaspores, after meiosis

2, 4 and 8-nucleate stages of the embryo sac

The mature embryo sac, 7-celled and 8-nucleate

Figure 1.H (a) Parts of the ovule showing a large megaspore mother cell, a dyad and a tetrad of megaspores; (b) 2, 4, and 8-nucleate stages of embryo sac and a mature embryo sac; (c) A diagrammatic representation of the mature embryo sac.

1.10 The female gametophyte

Meiosis in the MMC

- the MMC undergoes meiotic division, which leaves the importance of that meiosis to be worked out
- meiosis results in the production of four megaspores

Monosporic development

- in a majority of flowering plants one of the megaspores is functional while the other three degenerate
- only the functional megaspore develops into the female gametophyte, that is the embryo sac
- this method of embryo sac formation from a single megaspore is termed monosporic development, which leaves the ploidy of the cells of the nucellus, of the MMC, of the functional megaspore and of the female gametophyte to be worked out

1.11 The embryo sac

Free nuclear divisions

- the nucleus of the functional megaspore divides mitotically to form two nuclei, which move to opposite poles, forming the 2-nucleate embryo sac
- two more sequential mitotic nuclear divisions result in the 4-nucleate and later the 8-nucleate stages
- these mitotic divisions are strictly free nuclear, that is, nuclear divisions are not followed immediately by cell wall formation
- after the 8-nucleate stage cell walls are laid down, leading to the organisation of the typical female gametophyte or embryo sac, six of the eight nuclei being surrounded by cell walls and organised into cells while the remaining two, the polar nuclei, sit below the egg apparatus in the large central cell

How the cells are arranged

- three cells are grouped together at the micropylar end and constitute the egg apparatus, which consists of two synergids and one egg cell
- the synergids have special cellular thickenings at the micropylar tip called the filiform apparatus, which plays an important role in guiding the pollen tubes into the synergid
- three cells are at the chalazal end and are called the antipodals, and the large central cell has the two polar nuclei
- so a typical angiosperm embryo sac at maturity, though 8-nucleate, is 7-celled

1.12 Pollination

Why it is needed

- the male and female gametes in flowering plants are produced in the pollen grain and the embryo sac respectively, and as both types of gamete are non-motile they have to be brought together for fertilisation to occur
- pollination is the mechanism to achieve this objective, being the transfer of pollen grains shed from the anther to the stigma of a pistil
- flowering plants have evolved an amazing array of adaptations to achieve pollination, making use of external agents, which leaves the possible external agents to be listed

Autogamy

- in this type pollination is achieved within the same flower, the transfer of pollen grains being from the anther to the stigma of that same flower
- in a normal flower which opens and exposes the anthers and the stigma, complete autogamy is rather rare
- autogamy in such flowers requires synchrony in pollen release and stigma receptivity, and also that the anthers and the stigma lie close to each other so that self-pollination can occur

Self-pollinated flowers, pollen reaching their own stigma

Cross pollinated flowers, pollen carried between plants

Cleistogamous flowers, which never open at all

Figure 1.I (a) Self-pollinated flowers; (b) Cross pollinated flowers; (c) Cleistogamous flowers

1.13 Kinds of pollination

Cleistogamy

- some plants such as Viola or common pansy, Oxalis and Commelina produce two types of flower, chasmogamous flowers, which are similar to those of other species with exposed anthers and stigma, and cleistogamous flowers, which do not open at all
- in cleistogamous flowers the anthers and stigma lie close to each other, and when the anthers dehisce in the flower buds the pollen grains come in contact with the stigma to effect pollination
- such flowers are invariably autogamous, as there is no chance of cross-pollen landing on the stigma, and they produce assured seed-set even in the absence of pollinators, which leaves the question of whether cleistogamy is advantageous or disadvantageous to be argued

Geitonogamy and xenogamy

- geitonogamy is the transfer of pollen grains from the anther to the stigma of another flower of the same plant, and although it is functionally cross-pollination involving a pollinating agent, genetically it is similar to autogamy since the pollen grains come from the same plant
- xenogamy is the transfer of pollen grains from the anther to the stigma of a different plant
- xenogamy is the only type of pollination which during pollination brings genetically different types of pollen grain to the stigma

1.14 Agents of pollination

Abiotic and biotic

- plants use two abiotic agents, wind and water, and one biotic agent, animals
- the majority of plants use biotic agents, and only a small proportion use abiotic ones
- pollen grains coming in contact with the stigma is a chance factor in both wind and water pollination, so to compensate for this uncertainty and the associated loss of pollen grains the flowers produce an enormous amount of pollen compared with the number of ovules available for pollination

Compact inflorescence, many small flowers together

Well-exposed stamens, freeing pollen to the wind

Feathery stigma, trapping air-borne grains

Figure 1.J A wind-pollinated plant showing compact inflorecence and well-exposed stamens

1.15 Wind and water as agents

Pollination by wind

- pollination by wind is more common amongst abiotic pollinations
- it requires that the pollen grains are light and non-sticky so that they can be transported in wind currents
- such plants often possess well-exposed stamens, so that the pollen is easily dispersed into wind currents, and a large often-feathery stigma to easily trap air-borne pollen grains
- wind-pollinated flowers often have a single ovule in each ovary and numerous flowers packed into an inflorescence, a familiar example being the corn cob, whose ears are nothing but the stigma and style waving in the wind to trap pollen grains, and wind-pollination is quite common in grasses

Pollination by water

- it is quite rare in flowering plants and is limited to about 30 genera, mostly monocotyledons, whereas water is a regular mode of transport for the male gametes among the lower plant groups such as algae, bryophytes and pteridophytes
- it is believed, particularly for some bryophytes and pteridophytes, that their distribution is limited because of the need for water for the transport of male gametes and fertilisation
- some examples of water pollinated plants are Vallisneria and Hydrilla, which grow in fresh water, and several marine sea-grasses such as Zostera, while in a majority of aquatic plants such as water hyacinth and water lily the flowers emerge above the level of water and are pollinated by insects or wind as in most land plants

Two patterns in water

- in Vallisneria the female flower reaches the surface of the water by a long stalk and the male flowers or pollen grains are released on to the surface, to be carried passively by water currents until some of them eventually reach the female flowers and the stigma
- in another group such as the seagrasses, female flowers remain submerged and the pollen grains are released inside the water, the grains in many such species being long and ribbon like and carried passively inside the water until some reach the stigma
- in most water-pollinated species pollen grains are protected from wetting by a mucilaginous covering, and both wind and water pollinated flowers are not very colourful and do not produce nectar, which leaves the reason for that to be worked out

Vallisneria, its male flowers drifting on the surface

Female flower reached by them at the water line

An insect at a flower, dusted with pollen

The same insect touching the stigma of the next

Figure 1.K (a) Pollination by water in Vallisneria; (b) Insect pollination

1.16 Pollination by animals

Who the pollinators are

- the majority of flowering plants use a range of animals as pollinating agents, the common ones being bees, butterflies, flies, beetles, wasps, ants, moths, birds such as sunbirds and humming birds, and bats
- among the animals, insects and particularly bees are the dominant biotic pollinating agents
- even larger animals such as some primates, that is lemurs, arboreal or tree-dwelling rodents, and even reptiles such as the gecko lizard and the garden lizard have been reported as pollinators in some species
- flowers of animal-pollinated plants are often specifically adapted for a particular species of animal

How the flowers attract them

- the majority of insect-pollinated flowers are large, colourful, fragrant and rich in nectar, and when the flowers are small a number of them are clustered into an inflorescence to make them conspicuous
- animals are attracted to flowers by colour and/or fragrance, and the flowers pollinated by flies and beetles secrete foul odours to attract these animals
- to sustain animal visits the flowers have to provide rewards, nectar and pollen grains being the usual floral rewards
- for harvesting the rewards the animal visitor comes in contact with the anthers and the stigma, its body gets a coating of pollen grains, which are generally sticky in animal-pollinated flowers, and when it then comes in contact with a stigma it brings about pollination

A place to lay eggs

- in some species the floral reward is a safe place to lay eggs
- an example is the tallest flower, that of Amorphophallus, the flower itself being about 6 feet in height

Yucca and its moth

- a similar relationship exists between a species of moth and the plant Yucca, where both species cannot complete their life cycles without each other
- the moth deposits its eggs in the locule of the ovary and the flower in turn gets pollinated by the moth
- the larvae of the moth come out of the eggs as the seeds start developing

Watching pollinators for yourself

- observing the flowers of plants such as cucumber, mango, peepal, coriander, papaya, onion, lobia, cotton, tobacco, rose, lemon, eucalyptus and banana over a few days and at different times of the day shows which animals visit them and whether they could be pollinators
- it also shows whether there is any correlation between the characteristics of a flower and the animal that visits it, and only visitors that come in contact with the anthers and the stigma can bring about pollination
- many insects may consume pollen or nectar without bringing about pollination, and such floral visitors are referred to as pollen or nectar robbers

1.17 Outbreeding devices

Discouraging self-pollination

- the majority of flowering plants produce hermaphrodite flowers, so pollen grains are likely to come in contact with the stigma of the same flower, and continued self-pollination results in inbreeding depression
- in some species pollen release and stigma receptivity are not synchronised, either the pollen being released before the stigma becomes receptive or the stigma becoming receptive much before the release of pollen
- in some other species the anther and stigma are placed at different positions so that the pollen cannot come in contact with the stigma of the same flower, and both these devices prevent autogamy
- the third device is self-incompatibility, a genetic mechanism that prevents self-pollen, from the same flower or from other flowers of the same plant, from fertilising the ovules, by inhibiting pollen germination or pollen tube growth in the pistil

Unisexual flowers

- another device to prevent self-pollination is the production of unisexual flowers
- if both male and female flowers are present on the same plant, as in castor and maize, that is monoecious, it prevents autogamy but not geitonogamy
- in several species such as papaya male and female flowers are present on different plants, that is each plant is either male or female, which is dioecy, and this condition prevents both autogamy and geitonogamy

Pollen-pistil interaction

- pollination does not guarantee the transfer of the right type of pollen, that is compatible pollen of the same species as the stigma, and often pollen of the wrong type, either from other species or from the same plant if it is self-incompatible, also lands on the stigma
- the pistil has the ability to recognise whether the pollen is of the right type, that is compatible, or of the wrong type, that is incompatible
- if it is of the right type the pistil accepts the pollen and promotes the post-pollination events that lead to fertilisation

Pollen grains germinating on the stigma

Pollen tubes growing down through the style

Path of the tube in longitudinal section of the pistil

Entry of the tube into a synergid, through the filiform apparatus

Discharge of the two male gametes, one to the egg, one to the central cell

Figure 1.L (a) Pollen grains germinating on the stigma; (b) Pollen tubes growing through the style; (c) L.S. of pistil showing path of pollen tube growth; (d) enlarged view of an egg apparatus showing entry of pollen tube into a synergid; (e) Discharge of male gametes into a synergid and the movements of the sperms, one into the egg and the other into the central cell

1.18 Recognition, acceptance and rejection

The dialogue between pollen and pistil

- if the pollen is of the wrong type the pistil rejects it, by preventing pollen germination on the stigma or pollen tube growth in the style
- the ability of the pistil to recognise the pollen, followed by its acceptance or rejection, is the result of a continuous dialogue between pollen grain and pistil
- this dialogue is mediated by chemical components of the pollen interacting with those of the pistil
- it is only in recent years that botanists have been able to identify some of the pollen and pistil components and the interactions leading to recognition, followed by acceptance or rejection

Germination on the stigma

- following compatible pollination the pollen grain germinates on the stigma to produce a pollen tube through one of the germ pores
- the contents of the pollen grain move into the pollen tube

1.19 The pollen tube

From stigma to ovule

- the pollen tube grows through the tissues of the stigma and style and reaches the ovary
- in plants where pollen grains are shed at the two-celled condition, that is a vegetative cell and a generative cell, the generative cell divides and forms the two male gametes during the growth of the pollen tube in the stigma, whereas in plants which shed pollen in the three-celled condition the pollen tubes carry the two male gametes from the beginning
- the pollen tube, after reaching the ovary, enters the ovule through the micropyle and then enters one of the synergids through the filiform apparatus, many recent studies having shown that the filiform apparatus at the micropylar part of the synergids guides that entry
- all these events, from pollen deposition on the stigma until the pollen tubes enter the ovule, are together referred to as pollen-pistil interaction, a dynamic process involving pollen recognition followed by promotion or inhibition of the pollen, and knowledge of it would help the plant breeder manipulate the interaction, even in incompatible pollinations, to get desired hybrids

Seeing germination for yourself

- some pollen from flowers such as pea, chickpea, Crotalaria, balsam and Vinca can be dusted on a glass slide containing a drop of sugar solution of about 10 per cent
- after about 15 to 30 minutes the slide is observed under the low power lens of the microscope
- pollen tubes are then likely to be seen coming out of the pollen grains

1.20 Emasculation and bagging

Artificial hybridisation

- a breeder is interested in crossing different species and often genera to combine desirable characters and produce commercially superior varieties, and artificial hybridisation is one of the major approaches of a crop improvement programme
- in such crossing experiments it is important to make sure that only the desired pollen grains are used for pollination and the stigma is protected from contamination by unwanted pollen, which is achieved by emasculation and bagging techniques
- if the female parent bears bisexual flowers, the removal of anthers from the flower bud before the anther dehisces, using a pair of forceps, is necessary, and this step is referred to as emasculation
- emasculated flowers have to be covered with a bag of suitable size, generally made of butter paper, to prevent contamination of the stigma with unwanted pollen, and this process is called bagging

Completing the cross

- when the stigma of a bagged flower attains receptivity, mature pollen grains collected from the anthers of the male parent are dusted on the stigma, the flowers are rebagged, and the fruits are allowed to develop
- if the female parent produces unisexual flowers there is no need for emasculation
- the female flower buds are bagged before the flowers open, and when the stigma becomes receptive pollination is carried out using the desired pollen and the flower is rebagged

Fertilised embryo sac, with the zygote at the micropylar end

Primary Endosperm Nucleus, triploid, in the central cell

Proembryo, the first stage of embryo development

Globular and heart-shaped stages, then the mature embryo

Figure 1.M (a) Fertilised embryo sac showing zygote and Primary Endosperm Nucleus (PEN); (b) Stages in embryo development in a dicot [shown in reduced size as compared to (a)]

1.21 Double fertilisation

Syngamy and triple fusion

- after entering one of the synergids the pollen tube releases the two male gametes into the cytoplasm of the synergid
- one male gamete moves towards the egg cell and fuses with its nucleus, completing syngamy, which results in a diploid cell, the zygote
- the other male gamete moves towards the two polar nuclei in the central cell and fuses with them to produce a triploid primary endosperm nucleus (PEN), and as this involves the fusion of three haploid nuclei it is termed triple fusion
- since two types of fusion, syngamy and triple fusion, take place in an embryo sac, the phenomenon is termed double fertilisation, an event unique to flowering plants, and the central cell after triple fusion becomes the primary endosperm cell (PEC) and develops into the endosperm, while the zygote develops into an embryo

1.22 Post-fertilisation structures and events

What follows

- the events of endosperm and embryo development, the maturation of ovules into seeds and of the ovary into fruit, are collectively termed post-fertilisation events
- endosperm development precedes embryo development
- the primary endosperm cell divides repeatedly and forms a triploid endosperm tissue

A typical dicot embryo, with two cotyledons

Epicotyl above the cotyledons, ending in the plumule

Hypocotyl below, ending in the radicle and root cap

A grass embryo, with one cotyledon, the scutellum

Figure 1.N (a) A typical dicot embryo; (b) L.S. of an embryo of grass

Free-nuclear and cellular endosperm

- the cells of the endosperm tissue are filled with reserve food materials and are used for the nutrition of the developing embryo
- in the most common type of endosperm development the PEN undergoes successive nuclear divisions to give rise to free nuclei, and this stage is called free-nuclear endosperm, after which cell wall formation occurs and the endosperm becomes cellular
- the number of free nuclei formed before cellularisation varies greatly, and the coconut water from a tender coconut is nothing but free-nuclear endosperm, made up of thousands of nuclei, while the surrounding white kernel is the cellular endosperm
- endosperm may be completely consumed by the developing embryo before seed maturation, as in pea, groundnut and beans, or may persist in the mature seed, as in castor and coconut, and be used up during seed germination

1.23 The embryo

Embryogeny

- the embryo develops at the micropylar end of the embryo sac where the zygote is situated
- most zygotes divide only after a certain amount of endosperm is formed, an adaptation to provide assured nutrition to the developing embryo
- though the seeds differ greatly, the early stages of embryo development, that is embryogeny, are similar in both monocotyledons and dicotyledons, the zygote giving rise to the proembryo and subsequently to the globular, heart-shaped and mature embryo

The dicot embryo

- a typical dicotyledonous embryo consists of an embryonal axis and two cotyledons
- the portion of the embryonal axis above the level of the cotyledons is the epicotyl, which terminates in the plumule or stem tip
- the cylindrical portion below the level of the cotyledons is the hypocotyl, which terminates at its lower end in the radicle or root tip
- the root tip is covered with a root cap

The monocot embryo

- embryos of monocotyledons possess only one cotyledon, and in the grass family that cotyledon is called the scutellum, situated towards one side, that is laterally, of the embryonal axis
- at its lower end the embryonal axis has the radical and root cap enclosed in an undifferentiated sheath called the coleorrhiza
- the portion of the embryonal axis above the level of attachment of the scutellum is the epicotyl, which has a shoot apex and a few leaf primordia enclosed in a hollow foliar structure, the coleoptile
- soaking a few seeds of wheat, maize, peas, chickpeas or ground nut in water overnight and then splitting them shows the various parts of the embryo and the seed

1.24 The seed

What a seed contains

- in angiosperms the seed is the final product of sexual reproduction, often described as a fertilised ovule, and seeds are formed inside fruits
- a seed typically consists of seed coats, cotyledons and an embryo axis, the cotyledons being simple structures, generally thick and swollen due to storage of food reserves as in legumes
- mature seeds may be non-albuminous, with no residual endosperm as it is completely consumed during embryo development, as in pea and groundnut, or albuminous, retaining a part of the endosperm as it is not completely used up, as in wheat, maize, barley and castor
- occasionally, in some seeds such as black pepper and beet, remnants of the nucellus are also persistent, and this residual persistent nucellus is the perisperm

Seed coat and dormancy

- integuments of ovules harden as tough protective seed coats, and the micropyle remains as a small pore in the seed coat, which facilitates the entry of oxygen and water into the seed during germination
- as the seed matures its water content is reduced and seeds become relatively dry, at 10-15 per cent moisture by mass, and the general metabolic activity of the embryo slows down
- the embryo may enter a state of inactivity called dormancy, or, if favourable conditions of adequate moisture, oxygen and suitable temperature are available, germinate

1.25 Fruit and false fruit

The ovary becomes the fruit

- as ovules mature into seeds the ovary develops into a fruit, the two transformations proceeding simultaneously, and the wall of the ovary develops into the wall of the fruit, the pericarp
- fruits may be fleshy as in guava, orange and mango, or dry as in groundnut and mustard, and many fruits have evolved mechanisms for the dispersal of seeds, which leaves any relationship between the number of ovules in an ovary and the number of seeds in a fruit to be worked out
- in most plants the other floral parts degenerate and fall off by the time the fruit develops from the ovary, but in a few species such as apple, strawberry and cashew the thalamus also contributes to fruit formation, and such fruits are called false fruits, while most fruits develop only from the ovary and are called true fruits

Structure of some seeds, seed coat, cotyledons and embryo axis

Albuminous and non-albuminous kinds side by side

False fruit of apple, the thalamus grown fleshy

False fruit of strawberry, formed the same way

Figure 1.O (a) Structure of some seeds. (b) False fruits of apple and strawberry

1.26 Parthenocarpy, and why seeds matter

Fruit without fertilisation

- although in most species fruits are the result of fertilisation, there are a few species in which fruits develop without it, and such fruits are called parthenocarpic fruits
- banana is one such example
- parthenocarpy can be induced through the application of growth hormones, and such fruits are seedless

The advantages of a seed

- since reproductive processes such as pollination and fertilisation are independent of water, seed formation is more dependable
- seeds have better adaptive strategies for dispersal to new habitats and help the species to colonise other areas
- as they have sufficient food reserves, young seedlings are nourished until they are capable of photosynthesis on their own, and the hard seed coat provides protection to the young embryo
- being products of sexual reproduction, they generate new genetic combinations leading to variations

1.27 Seed viability

Seed and agriculture

- seed is the basis of our agriculture, and the dehydration and dormancy of mature seeds are crucial for their storage, so that they can be used as food throughout the year and to raise a crop in the next season
- agriculture in the absence of seeds, or with seeds that germinate straight away soon after formation and cannot be stored, is worth imagining
- the period for which seeds remain alive after dispersal varies greatly, a few species losing viability within a few months while the seeds of a large number of species live for several years

The oldest viable seeds

- some seeds can remain alive for hundreds of years, and there are several records of very old yet viable seeds
- the oldest is that of a lupine, Lupinus arcticus, excavated from Arctic Tundra, whose seed germinated and flowered after an estimated record of 10,000 years of dormancy
- a recent record of a 2000 years old viable seed is of the date palm, Phoenix dactylifera, discovered during the archeological excavation at King Herod's palace near the Dead Sea

Reproductive capacity

- the enormous reproductive capacity of some flowering plants is worth comprehending by asking how many eggs are present in an embryo sac, how many embryo sacs in an ovule, how many ovules in an ovary, how many ovaries in a typical flower and how many flowers on a tree
- orchid fruits each contain thousands of tiny seeds, and similar is the case in the fruits of some parasitic species such as Orobanche and Striga
- the tiny seed of Ficus grows into a very large tree, and each Ficus tree produces billions of seeds, so it is worth asking what other example there is in which such a tiny structure produces such a large biomass over the years

1.28 Apomixis

Seeds without fertilisation

- a few flowering plants, such as some species of Asteraceae and grasses, have evolved a special mechanism to produce seeds without fertilisation, called apomixis, which is a form of asexual reproduction that mimics sexual reproduction
- there are several ways of development of apomictic seeds, and in some species the diploid egg cell is formed without reduction division and develops into the embryo without fertilisation

1.29 Polyembryony and hybrid seed

Polyembryony

- more often, as in many Citrus and Mango varieties, some of the nucellar cells surrounding the embryo sac start dividing, protrude into the embryo sac and develop into embryos, so that each ovule contains many embryos
- the occurrence of more than one embryo in a seed is referred to as polyembryony
- squeezing some seeds of orange shows the many embryos of different sizes and shapes in each seed, which leaves the genetic nature of apomictic embryos, and whether they can be called clones, to be worked out

Why apomixis matters to breeders

- hybrid varieties of several of our food and vegetable crops are being extensively cultivated, and the cultivation of hybrids has tremendously increased productivity
- one of the problems of hybrids is that hybrid seeds have to be produced every year, since if seeds collected from hybrids are sown the plants in the progeny will segregate and do not maintain hybrid characters, and the production of hybrid seeds is costly, so their cost becomes too expensive for the farmers
- if these hybrids are made into apomicts there is no segregation of characters in the hybrid progeny, so farmers can keep using the hybrid seeds to raise a new crop year after year without having to buy fresh seed
- because of the importance of apomixis in the hybrid seed industry, active research is going on in many laboratories around the world to understand its genetics and to transfer apomictic genes into hybrid varieties

1.30 Summary

The anther and the pollen grain

- flowers are the seat of sexual reproduction in angiosperms, the androecium of stamens representing the male reproductive organs and the gynoecium of pistils the female ones
- a typical anther is bilobed, dithecous and tetrasporangiate, pollen grains develop inside the microsporangia, and four wall layers, the epidermis, endothecium, middle layers and tapetum, surround the microsporangium, while cells of the sporogenous tissue lying in its centre undergo meiosis, that is microsporogenesis, to form tetrads of microspores which mature into pollen grains
- pollen grains represent the male gametophytic generation, have a two-layered wall, the outer exine of sporopollenin with germ pores and the inner intine, and may have two cells, a vegetative cell and a generative cell, or three cells, a vegetative cell and two male gametes, at the time of shedding

1.31 Summary, continued

The pistil and the embryo sac

- the pistil has three parts, the stigma, style and ovary, and ovules are present in the ovary
- the ovules have a stalk called the funicle, protective integuments and an opening called the micropyle, and the central tissue is the nucellus, in which the archesporium differentiates
- a cell of the archesporium, the megaspore mother cell, divides meiotically and one of the megaspores forms the embryo sac, that is the female gametophyte
- the mature embryo sac is 7-celled and 8-nucleate, with the egg apparatus of two synergids and an egg cell at the micropylar end, three antipodals at the chalazal end, and at the centre a large central cell with two polar nuclei

Pollination and fertilisation

- pollination is the mechanism to transfer pollen grains from the anther to the stigma, and pollinating agents are either abiotic, that is wind and water, or biotic, that is animals
- pollen-pistil interaction involves all events from the landing of pollen grains on the stigma until the pollen tube enters the embryo sac when the pollen is compatible, or pollen inhibition when it is incompatible
- following compatible pollination the pollen grain germinates on the stigma and the resulting pollen tube grows through the style, enters the ovule and finally discharges two male gametes in one of the synergids
- angiosperms exhibit double fertilisation because two fusion events occur in each embryo sac, syngamy and triple fusion, whose products are the diploid zygote and the triploid primary endosperm nucleus in the primary endosperm cell, the zygote developing into the embryo and the primary endosperm cell forming the endosperm tissue, endosperm formation always preceding embryo development

Embryo, seed and apomixis

- the developing embryo passes through the proembryo, globular and heart-shaped stages before maturation, a mature dicotyledonous embryo having two cotyledons and an embryonal axis with epicotyl and hypocotyl, while embryos of monocotyledons have a single cotyledon
- after fertilisation the ovary develops into fruit and the ovules develop into seeds
- a phenomenon called apomixis is found in some angiosperms, particularly in grasses, and results in the formation of seeds without fertilisation, apomicts having several advantages in horticulture and agriculture
- some angiosperms produce more than one embryo in their seed, a phenomenon called polyembryony

1.32 Exercises

From this chapter

1 Name the parts of an angiosperm flower in which development of male and female gametophyte take place.
2 Differentiate between microsporogenesis and megasporogenesis. Which type of cell division occurs during these events? Name the structures formed at the end of these two events.
3 Arrange the following terms in the correct developmental sequence: Pollen grain, sporogenous tissue, microspore tetrad, pollen mother cell, male gametes.
4 With a neat, labelled diagram, describe the parts of a typical angiosperm ovule.
5 What is meant by monosporic development of female gametophyte?
6 With a neat diagram explain the 7-celled, 8-nucleate nature of the female gametophyte.

1.33 Exercises, continued

From this chapter, continued

1 What are chasmogamous flowers? Can cross-pollination occur in cleistogamous flowers? Give reasons for your answer.
2 Mention two strategies evolved to prevent self-pollination in flowers.
3 What is self-incompatibility? Why does self-pollination not lead to seed formation in self-incompatible species?
4 What is bagging technique? How is it useful in a plant breeding programme?
5 What is triple fusion? Where and how does it take place? Name the nuclei involved in triple fusion.
6 Why do you think the zygote is dormant for sometime in a fertilised ovule?
7 Differentiate between: (a) hypocotyl and epicotyl; (b) coleoptile and coleorrhiza; (c) integument and testa; (d) perisperm and pericarp.
8 Why is apple called a false fruit? Which part(s) of the flower forms the fruit?
9 What is meant by emasculation? When and why does a plant breeder employ this technique?
10 If one can induce parthenocarpy through the application of growth substances, which fruits would you select to induce parthenocarpy and why?
11 Explain the role of tapetum in the formation of pollen-grain wall.
12 What is apomixis and what is its importance?